

AOUANOUK Slimane

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Available for a 6-month internship starting January 2027 in robotics or autonomous systems.

Graduate Engineering student at Arts et Métiers and incoming Research Master's (M2) student in Advanced Systems and Robotics (SAR) at Sorbonne Université. Background in **physics-based modeling**, **numerical simulation**, **CAD**, **CFD**, **experimental analysis**, **Python data analysis**, and **system-level engineering**. Hands-on experience developing **AI-driven pipelines** for mechanical and autonomous systems, with a focus on **robot control** and **learning-based methods**.

EDUCATION — Combined GPA : 3.92 / 4.00

Sorbonne Université — Paris, France

Sept 2026 – Aug 2027

Research Master's (M2) in Advanced Systems & Robotics (SAR)

Research-focused curriculum: Robot Manipulators, State Estimation & Identification, Nonlinear Control, Deep Learning, Mobile Robotics; **30 ECTS** of advanced coursework followed by a **6-month internship**

Arts et Métiers Institute of Technology (ENSAM) — Paris, France

Sept 2024 – Jul 2027

Combined BS × MS Engineering Degree — Member of ParisTech; rigorous multidisciplinary training supported by world-class research laboratories.

- **Ranked top 12% of cohort (142 / 1223 students)**
- **Relevant Coursework:** Mechanical Design and Structural Analysis, Solid Mechanics, Robotics and Mechatronics, Computer-Aided Design (CAD), Design for Manufacturing and Assembly (DFM/DFA), Engineering Materials, Experimental Mechanics and Validation Testing, Fluid Mechanics, Heat Transfer, Numerical Optimization

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems

Jan–Aug 2026

Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisor: Prof. Francisco Chinesta

- Developing an **autonomous multi-robot system** spanning physical hardware, onboard sensing and computing; designed **Euler–Lagrange trajectory optimization** for **10+ robots**, processing visual data at **30–60 FPS** and replanning collision-free paths in **<100 ms**
- Integrated **computer vision**, **obstacle avoidance**, **real-time planning**, and **distributed control**; integrating **AI and digital twins**; reduced solver runtime by **89.7%** versus baseline; publication in preparation

Construction Site Worker Intern — Société de Rénovation Parisienne (SRP)

Summer 2025

- Supported large-scale construction operations on a major renovation project involving **3,000+ tons of excavated soil**. Conducted **on-site measurements**, **markings**, and **consistency checks** under real-world tolerances during a **9–10 month underground phase**. Coordinated with subcontractors and adapted plans to field constraints, gaining hands-on exposure to **field data**, safety requirements, and execution–theory gaps

SELECTED PROJECTS

Miniature Formula One Car Design – Arts et Métiers

2025

- Designed a miniature F1 car optimized for a 20 m straight-line race; developed the full CAD model in **Fusion 360** with 3–4 design iterations; conducted **CFD simulations** in **STAR-CCM+** at 50 m/s to analyze pressure and drag; used simulation results to guide aerodynamic optimization; produced a 3D-printed prototype and prepared CNC machining while ensuring compliance with 10+ technical regulations

AI-Driven Smart Workshop Project – Arts et Métiers

2025

- Built an **end-to-end local data & AI pipeline** to query **technical workshop data** in **natural language**; structured **heterogeneous CSV data** and migrated it to a **relational SQLite database**; **validated and explored data** using **Python** and **pandas**; designed **SQL views** to simplify **complex joins** and improve **robustness**; implemented a **secure Text-to-SQL system** powered by a **locally deployed LLM (Ollama)** with **strict query constraints**; developed a **Streamlit interface** enabling **real-time usage** and **human-readable responses**

Aircraft Brake System Design Project – Arts et Métiers

2025

- Designed a **complete mechanical braking system** for a **light aircraft wheel** composed of **10+ components** using **Fusion 360**; built a **physics-based functional architecture** ensuring **torque transmission** across **5+ mechanical interfaces**; **quantitatively modeled load paths** and **mechanical constraints**; performed **engineering validations** including **bearing preload analysis**, **bolt slip criteria**, **shaft stress verification**, and **brake disc thermal dissipation**; ensured **system consistency** through **analytical checks** and **safety margins**

SKILLS AND INTERESTS

Data & Programming: Python (data analysis, numerical computation), SQLite, MATLAB, Arduino, LaTeX

Modeling & Simulation: Computational Fluid Dynamics (STAR-CCM+), Finite Element Analysis (Abaqus), Mechanical CAD (Fusion 360, 3DEXPERIENCE), Simulink

Experimental Robotics: Multi-robot hardware integration, onboard vision and sensing, real-time trajectory testing

Languages: French (native), English (TOEFL iBT: 5/6), Italian (basic), Spanish (basic)

Interests: Data-driven engineering, applied AI, basketball, soccer, strength training